Mars Habitat Automation Platform

Distributed Event-Driven System for Habitat Monitoring & Control

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Laboratory of Advanced Programming – Hackathon Exam 2025/2026

Agenda

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The Challenge

A Mars habitat's automation stack has partially failed, leaving devices operating with incompatible protocols.

Multiple Data Sources

REST sensors, SSE streams, WebSocket telemetry

Heterogeneous Formats

Different payload schemas requiring normalization

Real-time Requirements

Immediate response to environmental changes

Automation Needs

Rules-based actuator control for habitat safety

Our Solution

A distributed event-driven platform with unified monitoring and control

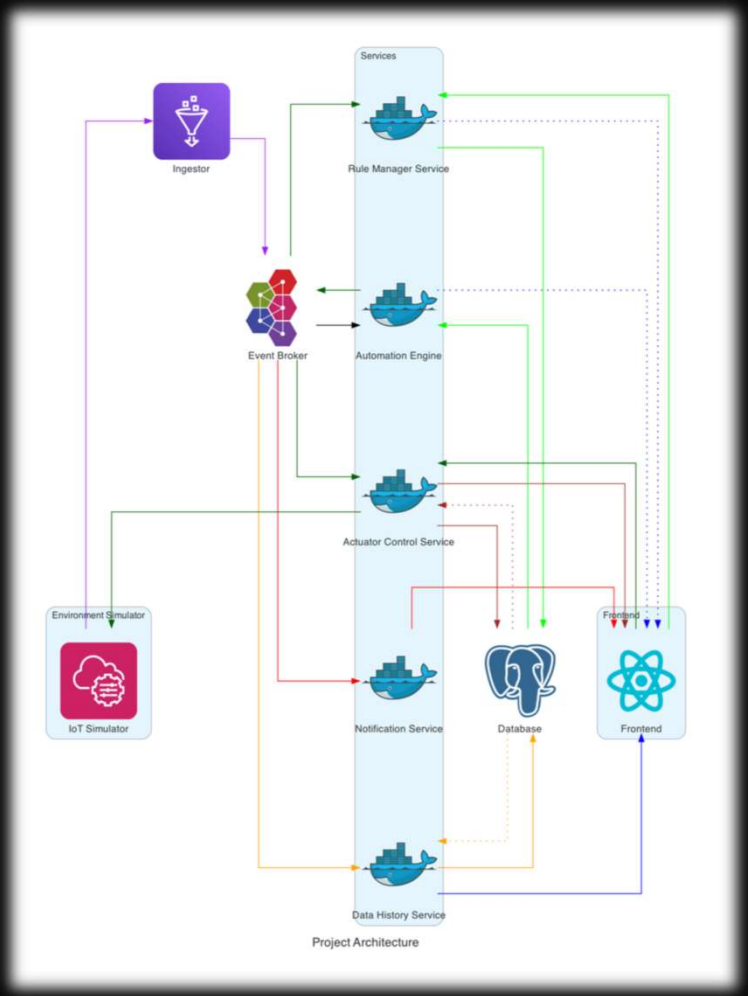
Unified Ingestion
REST polling + SSE streams normalized to single event schema

Event-Driven Core
RabbitMQ message broker for decoupled microservices

Smart Automation
IF-THEN rules with automatic actuator control

Real-Time Dashboard
React SPA with live updates and manual controls

System Architecture



Microservices Architecture

Ingestion
8001

REST polling & SSE streams
Schema normalization

Automation Engine
8002

Rule evaluation
Event-driven triggers

Rule Manager
8003

CRUD API for automation rules persistence

Notification
8004

Real-time SSE alerts
Event broadcasting

Actuator Control
8005

Device command execution
Audit logging

Data History
8006

Historical queries
In-memory cache

Infrastructure Services

RabbitMQ Message Broker

- AMQP protocol (port 5672)
- Topic exchange: mars_events
- Event-driven pub/sub architecture
- Durable queues with TTL
- Management UI on port 15672

PostgreSQL Database

- PostgreSQL 16 (port 5433)
- automation_rules table
- sensor_readings history
- actuator_commands audit
- Persistent volume storage

Unified Event Schema

{ event_id, sensor_id, timestamp, metric, value, unit, source, status, raw_schema }

All sensor data normalized to a single format • REST polls + SSE streams → unified events

Data Ingestion Service

REST Sensors (Polling)

- greenhouse_temperature
- entrance_humidity
- co2_hall
- hydroponic_ph
- water_tank_level
- corridor_pressure
- air_quality_pm25
- air_quality_voc

Telemetry Streams (SSE)

- solar_array
- radiation
- life_support
- thermal_loop
- power_bus
- power_consumption
- airlock

Normalization Pipeline

8 different REST schemas + 7 telemetry schemas → Single unified event format
Published to RabbitMQ with routing key: events.sensor.{sensor_id}

Automation Engine

Event-Driven Rule Processing

1. **Receive** — RabbitMQ event
2. **Parse** — Extract sensor_id
3. **Match** — Find rules
4. **Evaluate** — Check condition
5. **Execute** — Publish command

IF-THEN Rule Model

IF <sensor_id> <operator> <value> THEN set <actuator_id> to ON | OFF

Operators: < <= = > >= • Example: IF greenhouse_temperature > 28 THEN set cooling_fan to ON

Frontend Application

Technology Stack

- React 18 + TypeScript
- Vite 6 Build Tool
- React Router 7
- Tailwind CSS 4
- Recharts for Charts
- Radix UI Components

Application Pages

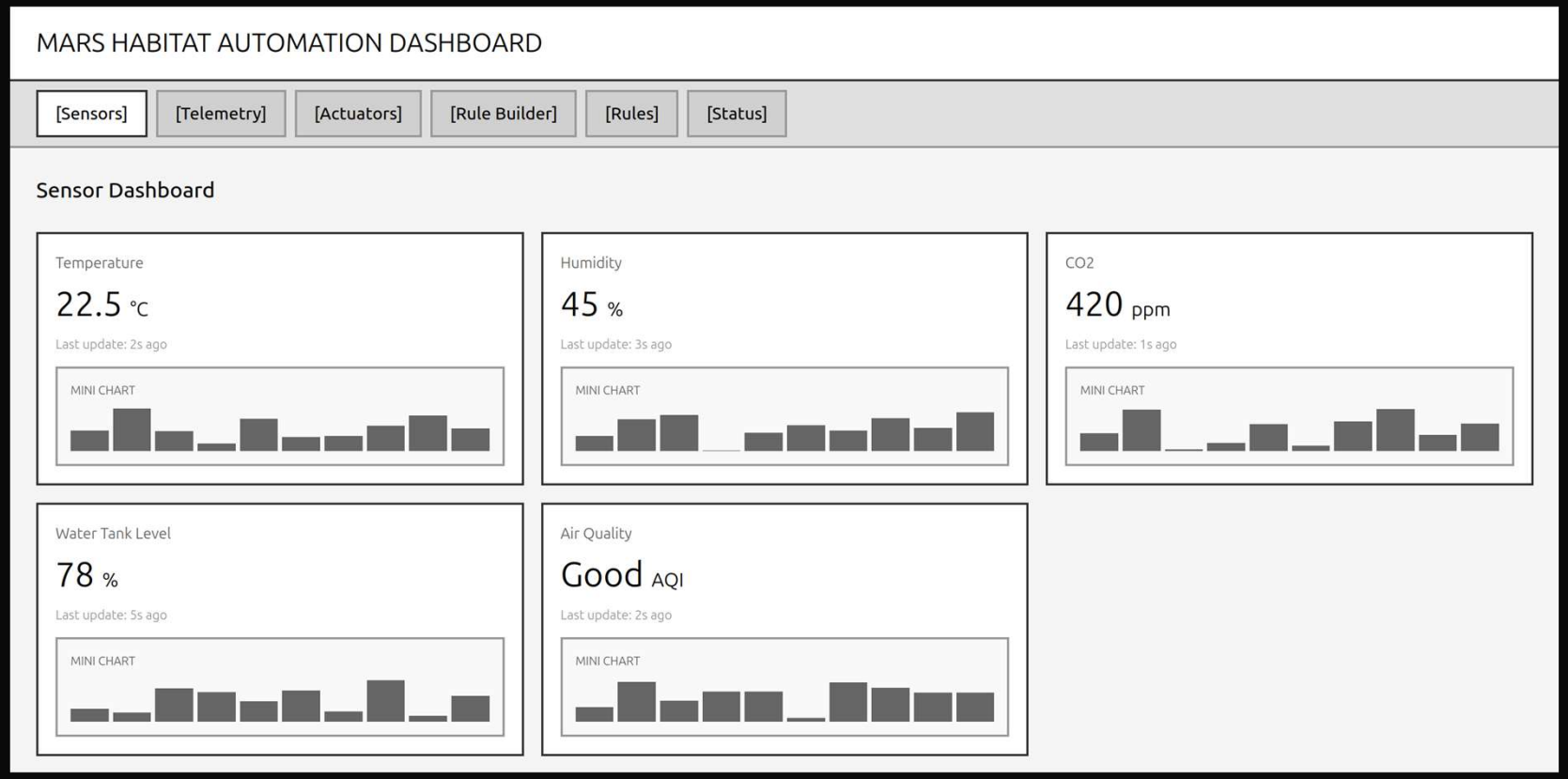
Sensors	REST sensors dashboard with real-time updates
Telemetry	Streaming data visualization with charts
Actuators	Device control and command history
Rule Builder	Create and edit automation rules
Rules	List, enable/disable, delete rules
Status	System health and notification log

Real-Time Features

Server-Sent Events (SSE) for live notifications • Aggressive polling for sensor updates
React state management for instant UI feedback • Recharts for live data visualization

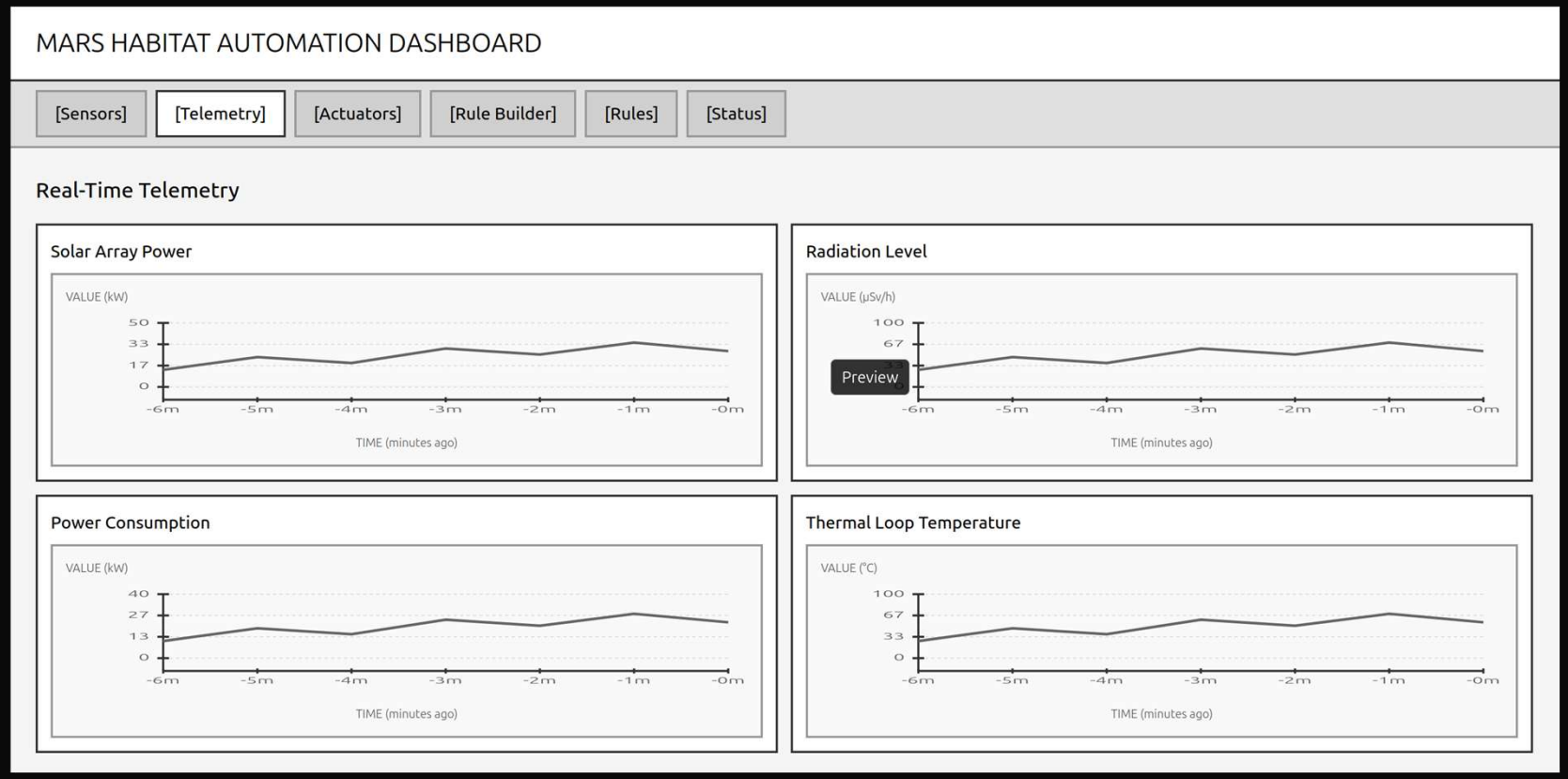
Sensors Dashboard

Real-time REST sensor monitoring with status indicators



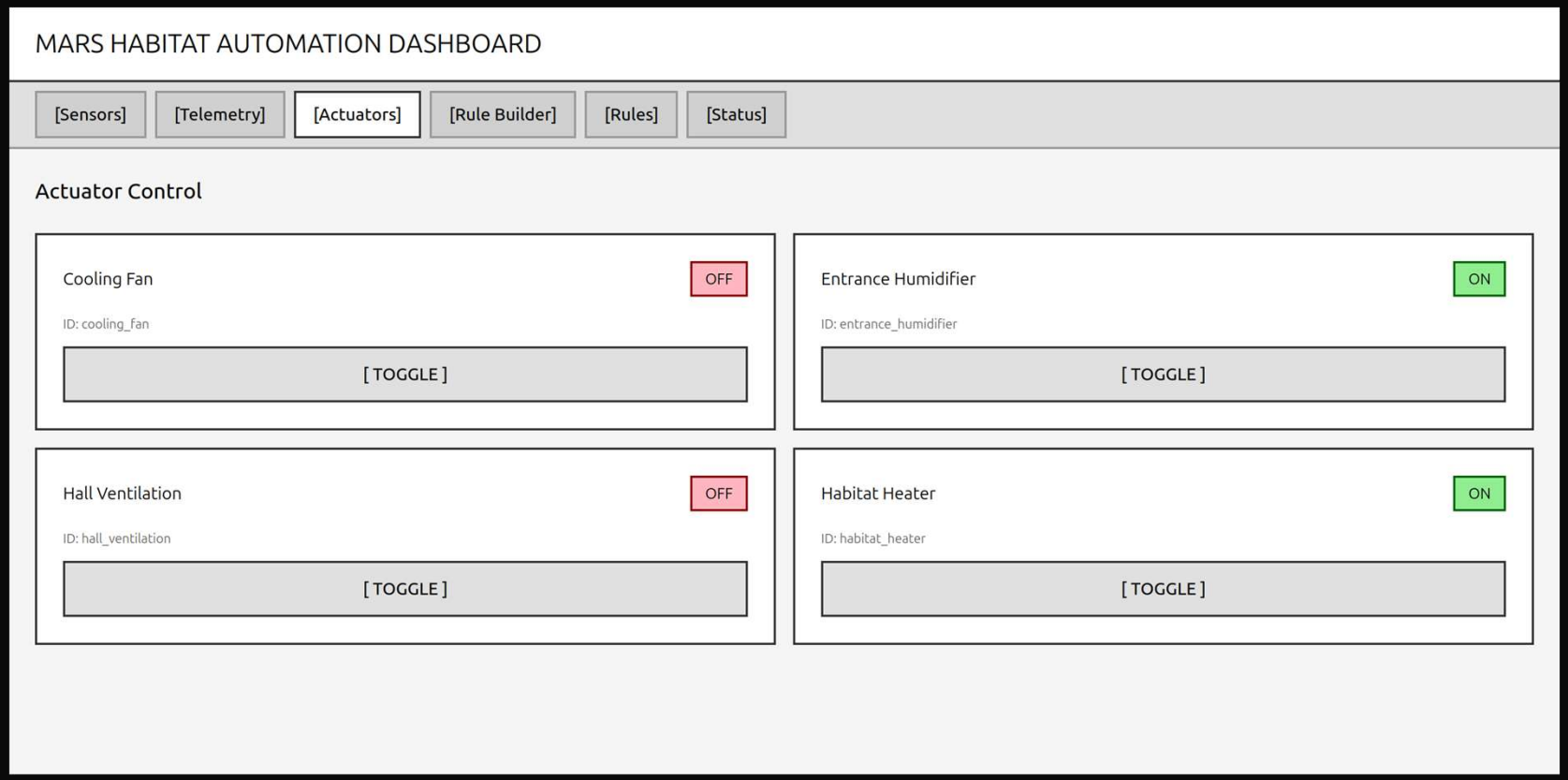
Telemetry Streaming

SSE-powered live data visualization



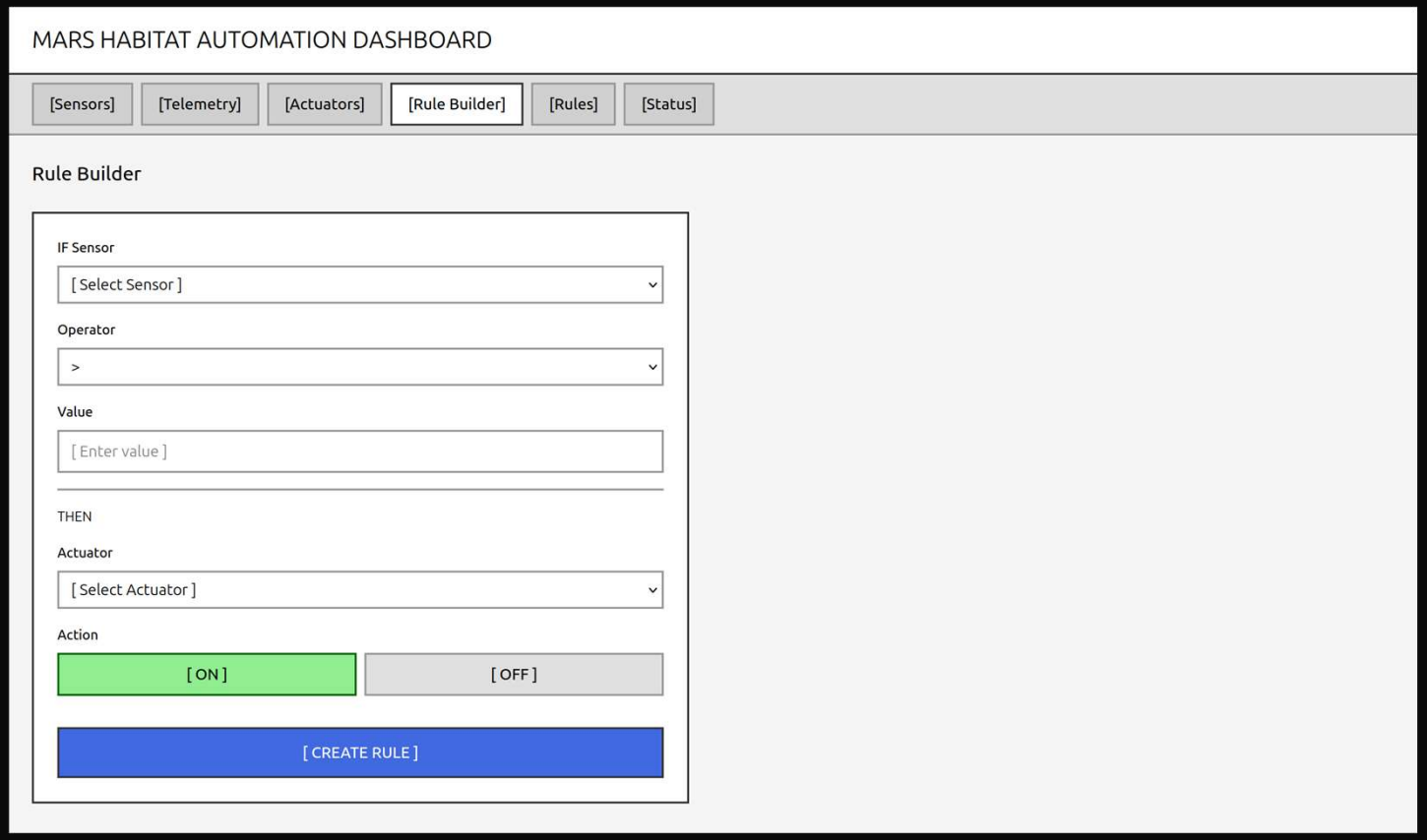
Actuator Control

Manual control and command history audit trail



Rule Builder

Create IF-THEN automation rules visually



Rules Management

List, toggle, edit, and delete automation rules

MARS HABITAT AUTOMATION DASHBOARD

[Sensors] [Telemetry] [Actuators] [Rule Builder] [Rules] [Status]

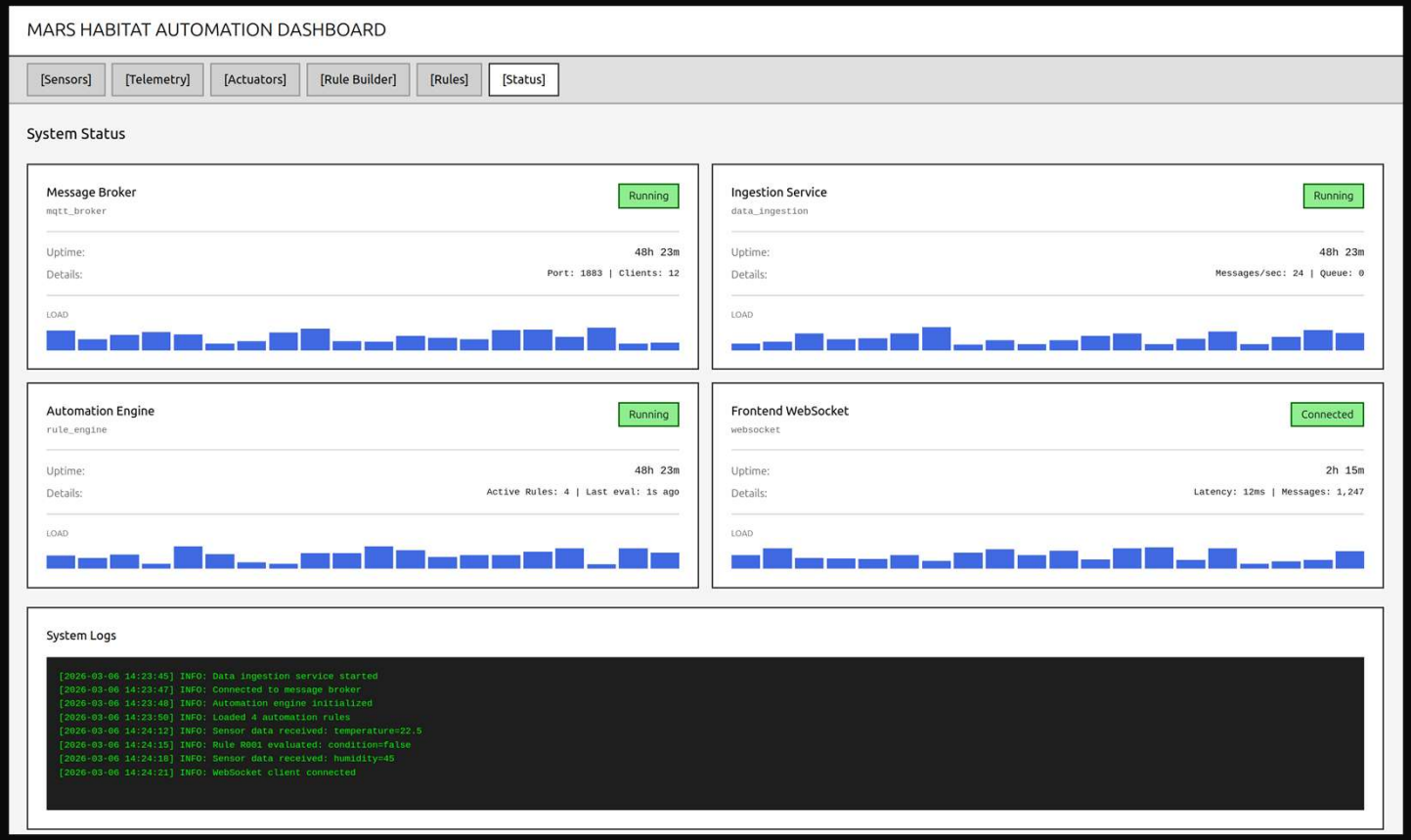
Active Automation Rules

Rule ID	Condition	Action	Status	Last Triggered
R001	temperature > 25	cooling_fan = ON	Active	2 min ago
R002	humidity < 40	entrance_humidifier = ON	Active	15 min ago
R003	co2 > 600	hall_ventilation = ON	Active	Never
R004	temperature < 18	habitat_heater = ON	Inactive	3 hours ago

[EDIT] [DELETE] [ENABLE/DISABLE]

System Status

Health monitoring and notification center



Database Design

automation_rules

- id (SERIAL PK)
- name, description
- sensor_id, operator
- threshold_value, threshold_unit
- actuator_id, actuator_action
- is_active (BOOLEAN)
- created_at, updated_at

sensor_readings

- id (BIGSERIAL PK)
- sensor_id (VARCHAR)
- value (DECIMAL)
- unit (VARCHAR)
- source (rest/stream)
- recorded_at (TIMESTAMP)
- created_at (TIMESTAMP)

actuator_commands

- id (BIGSERIAL PK)
- actuator_id (VARCHAR)
- previous_state, new_state
- source (auto/manual)
- reason (TEXT)
- rule_id (FK)
- executed_at (TIMESTAMP)

Data Flow Architecture

Event Routing Keys

Sensor Events:
events.sensor.{sensor_id}

Actuator Commands:
commands.actuator.{actuator_id}

Notifications:
events.#

Technology Stack

Backend Services

- Python 3.11+
- FastAPI (Async)
- aio-pika (RabbitMQ)
- asyncpg + SQLAlchemy
- httpx (HTTP Client)
- Pydantic (Validation)
- Uvicorn (ASGI)

Frontend

- React 18
- TypeScript
- Vite 6
- React Router 7
- Tailwind CSS 4
- Recharts
- Radix UI

Infrastructure

- PostgreSQL 16
- RabbitMQ (AMQP)
- Docker
- Docker Compose
- Nginx (Reverse Proxy)

Containerized Microservices with Event-Driven Architecture

Key User Stories

Monitoring

- Real-time sensor values
- Live telemetry from SSE
- Automatic dashboard updates
- Status indicators (ok/warning)
- Sensor filtering by category

Control

- View actuator ON/OFF states
- Manual actuator toggle
- Automatic rule-based control
- Command history audit

Automation

- Create IF-THEN rules
- View all automation rules
- Edit existing rules
- Delete/enable/disable
- Real-time trigger alerts

Analytics

- Live time charts
- Historical data queries
- System health summary
- Connectivity monitoring

Conclusions

Project Achievements

- Distributed microservices architecture with event-driven communication
- Real-time monitoring and control of simulated Mars habitat
- IF-THEN automation rules with automatic actuator control
- Modern React dashboard with SSE-based live updates
- Containerized deployment with Docker Compose
- Unified event schema for heterogeneous data sources

Future Enhancements

→ **MachineLearning:** Predictive maintenance and anomaly detection
→ **Mobile App:** Remote monitoring and push notifications
→ **Advanced Alerting:** Email/SMS notifications for critical events

Thank You

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